

LETHBRIDGE DEMO FARM AGDM, AB, Canada

WMO: 715090

Lat: 49.6867N

Lon: 112.7449W

Elev: 2975

StdP: 13.18

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.6	-11.6	-23.2	1.7	-17.3	-16.9	2.4	-11.3	37.4	40.4	34.8	40.3	6.0	0	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.8	86.2	61.9	83.1	61.1	80.2	60.2	66.6	78.6	64.4	77.3	62.4	75.8	9.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	92.8	73.7	59.6	85.0	70.9	57.2	78.0	68.3	33.0	78.6	31.1	77.3	29.5	76.3	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.7	28.0	24.8	DB	-28.3	92.6	5.4	3.3	-32.2	95.0	-35.4	97.0	-38.4	98.8	-42.3	101.3	(4)
(5)			WB	-27.9	71.6	5.8	2.8	-32.0	73.6	-35.4	75.2	-38.6	76.8	-42.8	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	24.5	22.4	32.7	42.1	52.1	59.0	64.8	62.8	55.1	43.6	32.4	21.8	(6)
(7)		DBStd	18.73	15.41	15.65	12.94	8.93	7.71	5.53	4.80	5.59	8.03	9.45	13.20	15.08	(7)
(8)		HDD50	4118	791	774	540	259	67	3	0	1	41	235	533	874	(8)
(9)		HDD65	8192	1255	1194	1003	688	403	194	63	106	305	663	979	1339	(9)
(10)		CDD50	1525	1	1	3	21	133	272	460	399	194	37	4	0	(10)
(11)		CDD65	122	0	0	0	0	3	13	58	39	9	0	0	0	(11)
(12)		CDH74	2551	0	0	0	17	162	254	979	807	324	8	0	0	(12)
(13)		CDH80	678	0	0	0	1	28	47	272	245	85	0	0	0	(13)
(14)	Wind	WSAvg	10.4	12.8	10.3	10.9	11.3	10.1	9.8	8.0	7.6	8.8	10.6	12.3	11.8	(14)
(15)	Precipitation	PrecAvg	16.7	0.9	0.6	1.1	1.6	2.1	2.6	1.8	1.7	1.8	0.7	0.7	0.9	(15)
(16)		PrecMax	30.2	2.3	1.5	3.1	6.7	5.3	6.7	4.8	5.6	5.5	1.8	2.0	2.3	(16)
(17)		PrecMin	9.6	0.2	0.1	0.2	0.0	0.4	0.1	0.3	0.2	0.2	0.0	0.0	0.2	(17)
(18)		PrecStd	4.4	0.5	0.3	0.6	1.3	1.2	1.7	1.3	1.3	1.4	0.5	0.6	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.0	56.9	63.8	76.4	83.9	86.0	90.4	90.6	87.1	74.2	64.5	53.6	(19)
(20)			MCWB	43.8	41.8	46.4	52.2	56.9	63.2	63.7	61.3	59.0	53.2	48.3	41.6	(20)
(21)		2%	DB	50.9	51.2	59.0	68.6	78.8	80.4	86.0	85.8	82.2	68.3	57.4	49.0	(21)
(22)			MCWB	41.1	39.7	44.6	48.9	55.9	60.2	64.3	61.0	57.6	51.4	44.3	40.0	(22)
(23)		5%	DB	46.8	46.9	55.0	63.4	73.8	76.6	83.0	82.4	77.6	63.4	53.1	45.3	(23)
(24)			MCWB	38.3	37.3	42.3	46.4	53.7	59.6	63.7	60.3	56.2	49.1	42.6	37.5	(24)
(25)		10%	DB	43.5	43.0	50.7	58.4	69.1	73.0	79.8	78.6	71.9	58.8	49.3	42.2	(25)
(26)			MCWB	36.3	35.0	40.1	44.1	51.7	57.2	62.5	59.6	54.5	46.6	40.1	35.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	43.9	48.1	53.1	58.6	67.4	71.2	66.9	62.0	55.4	49.4	43.6	(27)
(28)			MCD B	54.3	53.6	60.9	73.1	77.5	79.4	83.6	79.6	78.8	69.9	62.8	52.0	(28)
(29)		2%	WB	41.1	40.4	45.3	49.8	56.4	63.5	68.1	64.4	59.1	52.5	45.8	40.5	(29)
(30)			MCD B	50.1	50.2	57.2	67.4	75.4	76.1	80.6	79.3	77.1	65.7	56.3	47.9	(30)
(31)		5%	WB	38.7	37.6	42.8	47.1	54.5	60.8	65.7	62.5	57.2	49.7	43.0	37.9	(31)
(32)			MCD B	46.3	46.3	53.8	61.5	71.2	72.9	77.6	77.1	75.3	61.8	52.1	44.5	(32)
(33)		10%	WB	36.6	35.3	40.4	44.7	52.7	58.7	63.8	60.7	55.0	47.2	40.6	35.5	(33)
(34)			MCD B	43.1	42.6	50.1	57.3	66.9	70.9	76.3	74.9	70.9	57.7	48.9	42.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.0	20.2	20.9	22.7	23.5	21.8	25.8	26.3	25.2	21.9	19.6	18.3	(35)
(36)			MCDBR	19.5	24.0	27.8	32.7	33.1	29.6	32.6	35.3	36.5	29.0	24.2	19.1	(36)
(37)			MCWBR	12.5	14.8	15.8	16.4	14.2	13.8	15.4	14.8	15.4	15.0	14.3	12.7	(37)
(38)		5% WB	MCD BR	19.4	23.7	26.8	30.7	29.8	26.4	28.3	29.9	33.6	28.0	23.4	19.7	(38)
(39)			MCWBR	13.1	15.4	16.0	16.0	13.5	13.5	15.5	14.2	14.7	14.9	14.4	13.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.251	0.259	0.284	0.320	0.333	0.330	0.333	0.350	0.312	0.280	0.256	0.234	(40)	
(41)		taud	2.459	2.491	2.483	2.391	2.406	2.460	2.467	2.395	2.473	2.541	2.505	2.458	(41)	
(42)		Ebn at Noon	264	289	298	294	293	294	292	281	283	276	259	255	(42)	
(43)		Edh at Noon	18	23	28	34	35	34	33	34	28	22	17	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	393	710	1111	1536	1830	2036	2237	1778	1263	786	439	310	(44)	
(45)		RadStd	32	56	92	103	152	147	115	109	106	80	35	30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page